

time-domain metrics

rise time: first reach of final value

$$t_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}}$$

$$\sigma\% = e^{-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}} \times 100\%$$

settling-time approximations

$$t_s(5\%) \approx \frac{3}{\zeta\omega_n}$$

$$t_s(2\%) \approx \frac{4}{\zeta\omega_n}$$

best damping ratio: $\zeta \approx 0.707$